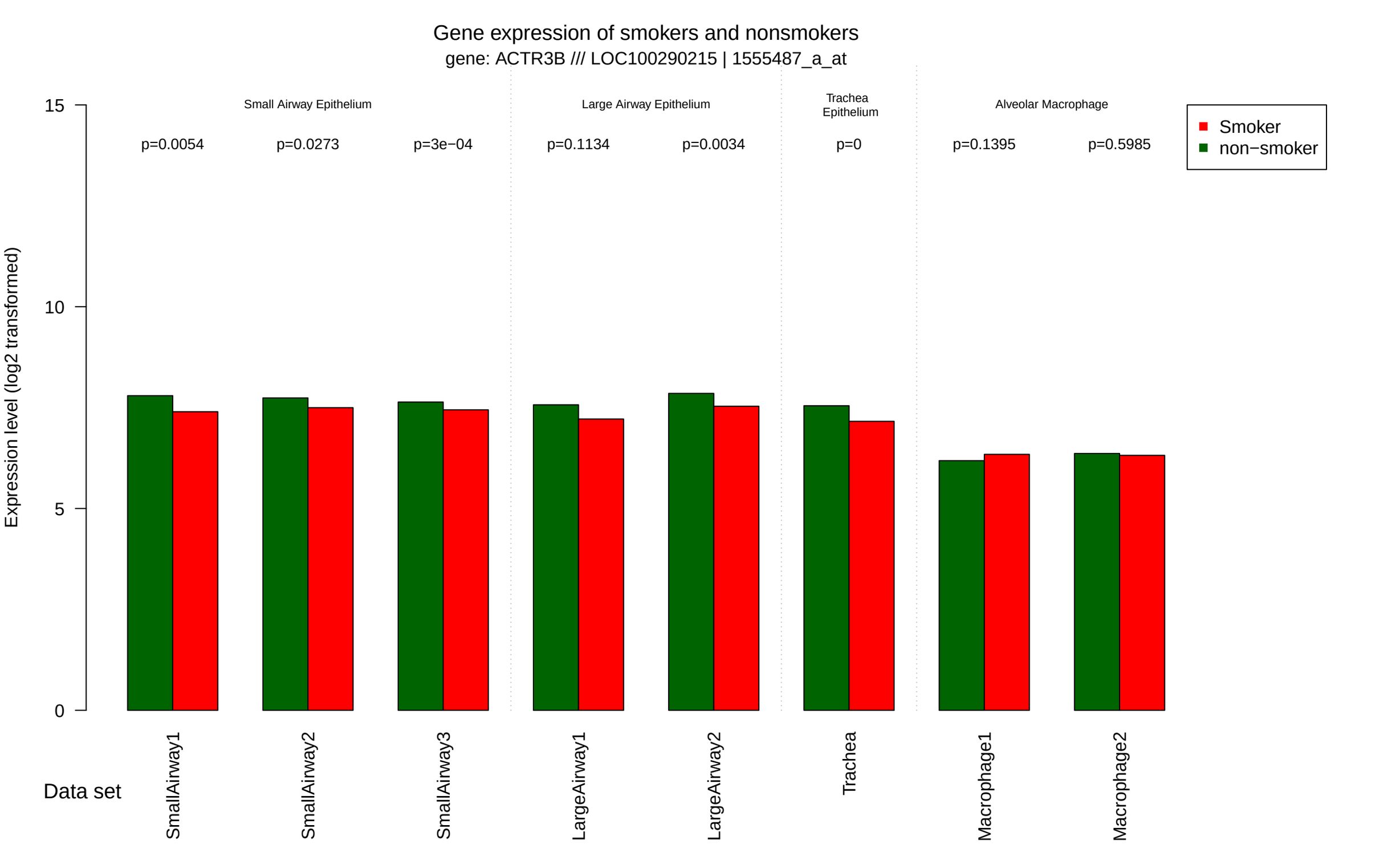
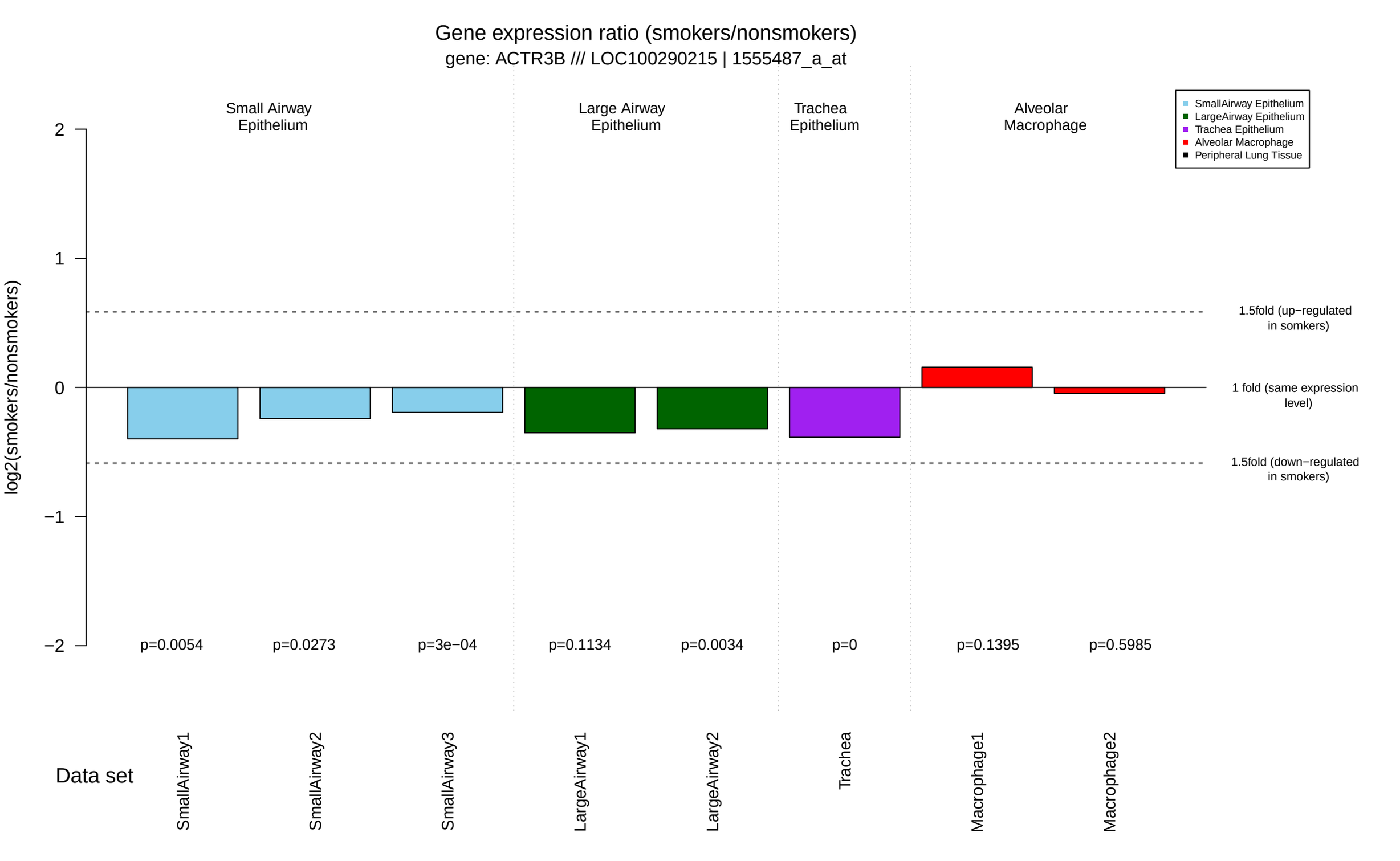
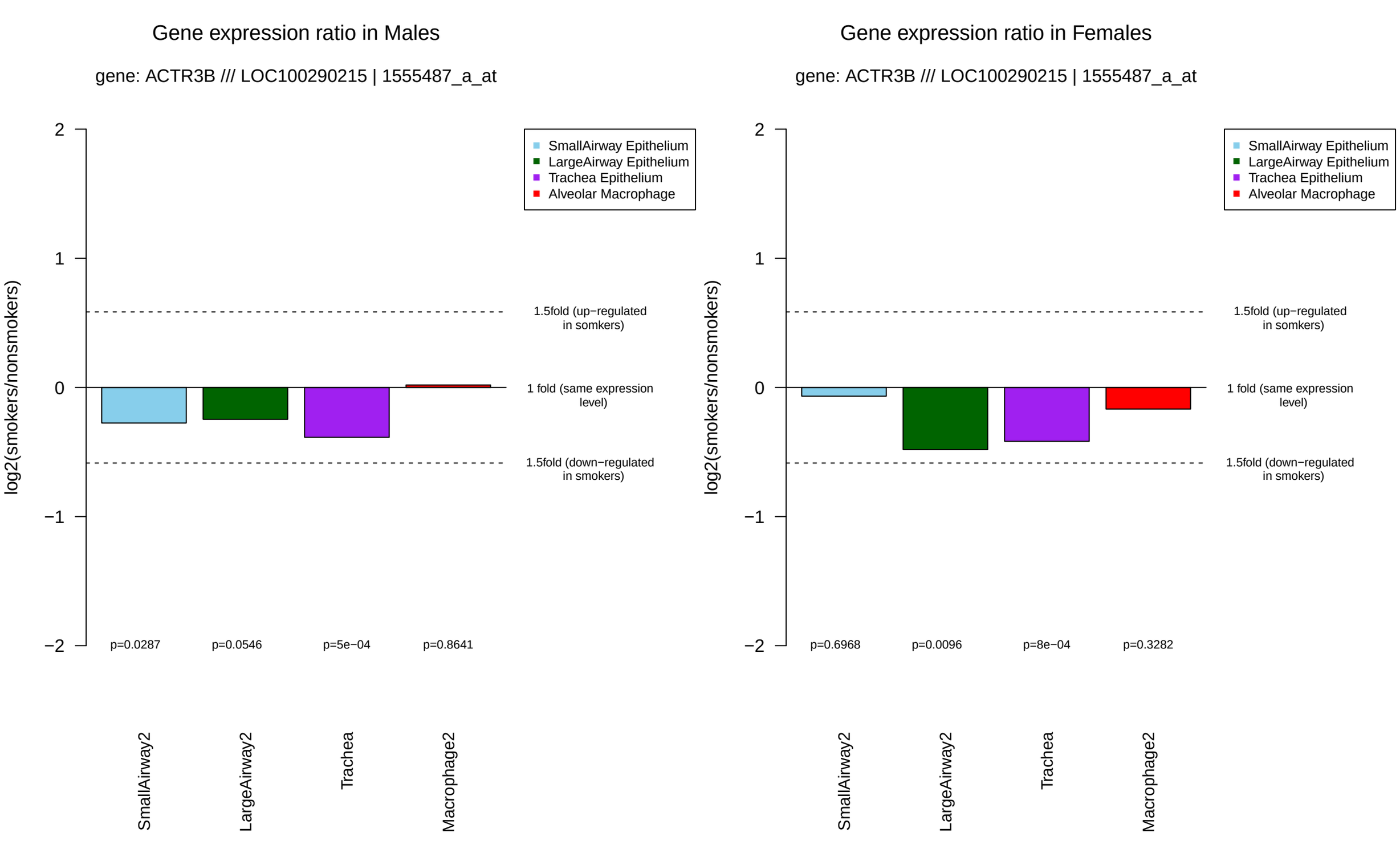
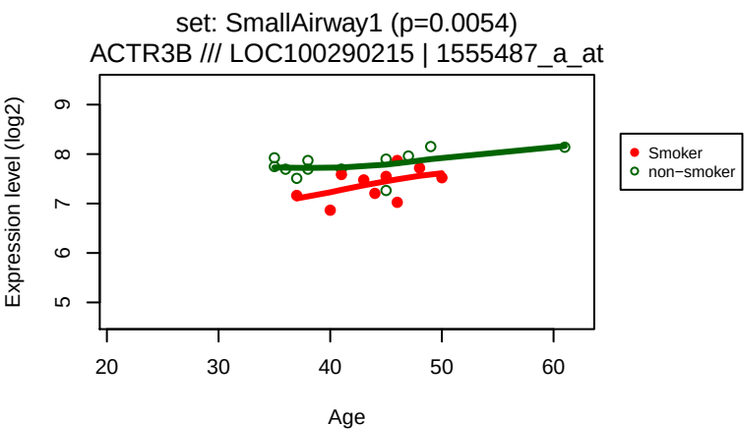
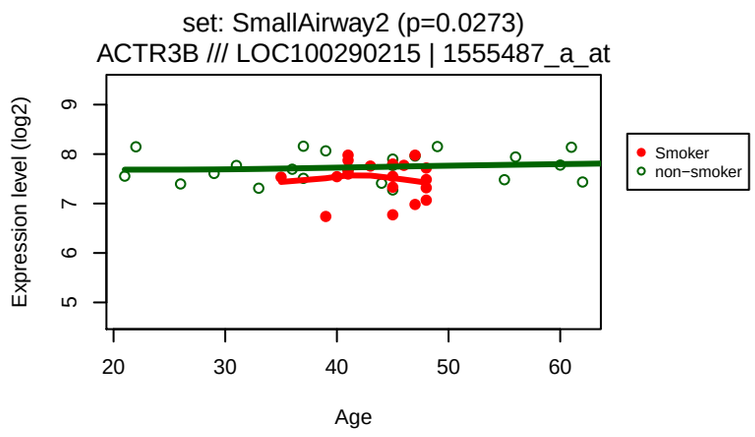
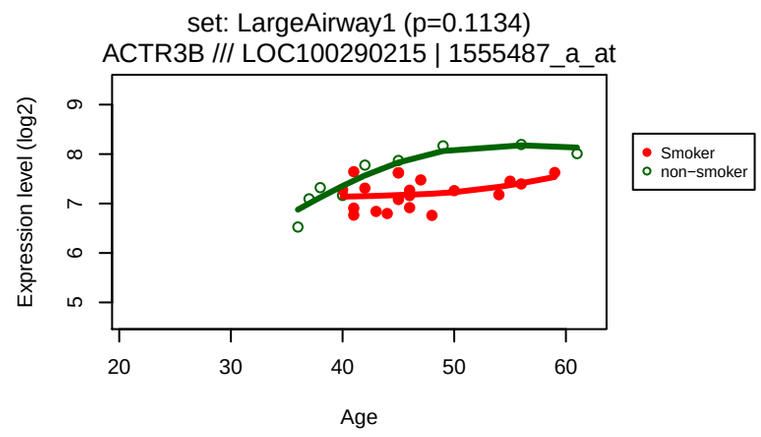

Gene expression of smokers and nonsmokers
gene: ACTR3B /// LOC100290215 | 1555487_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0054
p=0.0273
p=3e−04
p=0.1134
p=0.0034
p=0
p=0.1395
p=0.5985
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

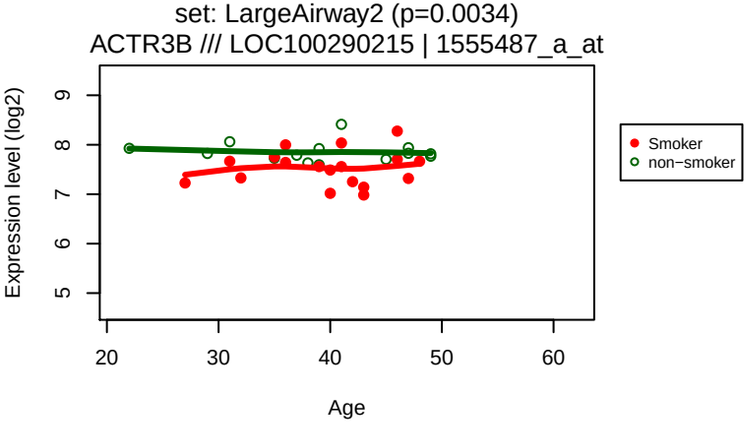

Gene expression ratio (smokers/nonsmokers)
gene: ACTR3B /// LOC100290215 | 1555487_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0054
p=0.0273
p=3e−04
p=0.1134
p=0.0034
p=0
p=0.1395
p=0.5985
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

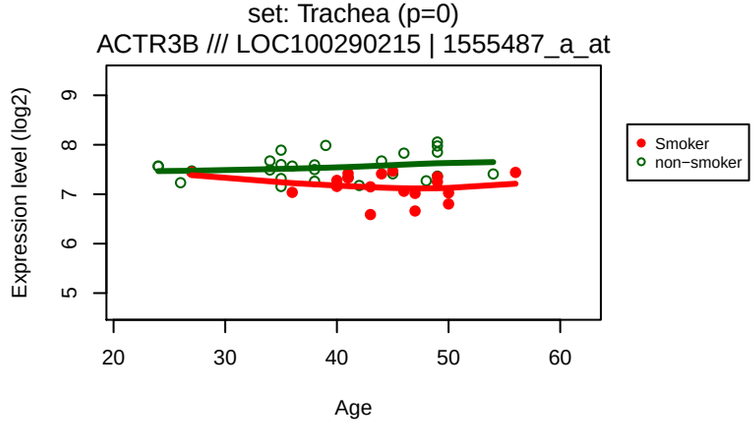

Gene expression ratio in Males
gene: ACTR3B /// LOC100290215 | 1555487_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0287
p=0.0546
p=5e−04
p=0.8641
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR3B /// LOC100290215 | 1555487_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6968
p=0.0096
p=8e−04
p=0.3282
SmallAirway2
LargeAirway2
Trachea
Macrophage2

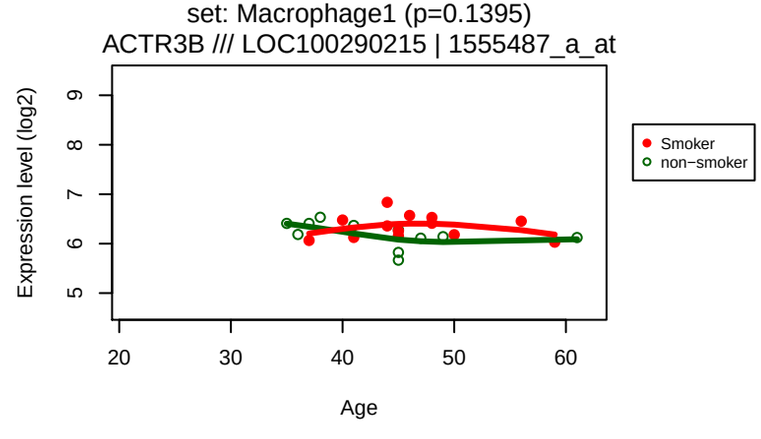

set: SmallAirway1 (p=0.0054)
ACTR3B /// LOC100290215 | 1555487_a_at
Expression level (log2)
Age
Smoker
non−smoker

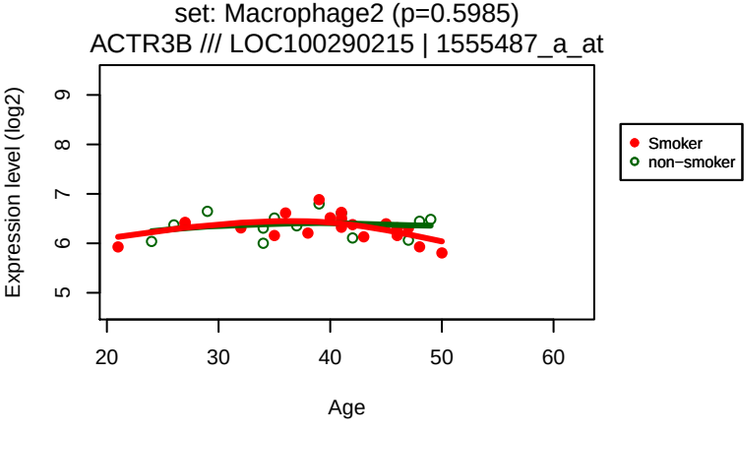

set: SmallAirway2 (p=0.0273)
ACTR3B /// LOC100290215 | 1555487_a_at
Expression level (log2)
Age
Smoker
non−smoker

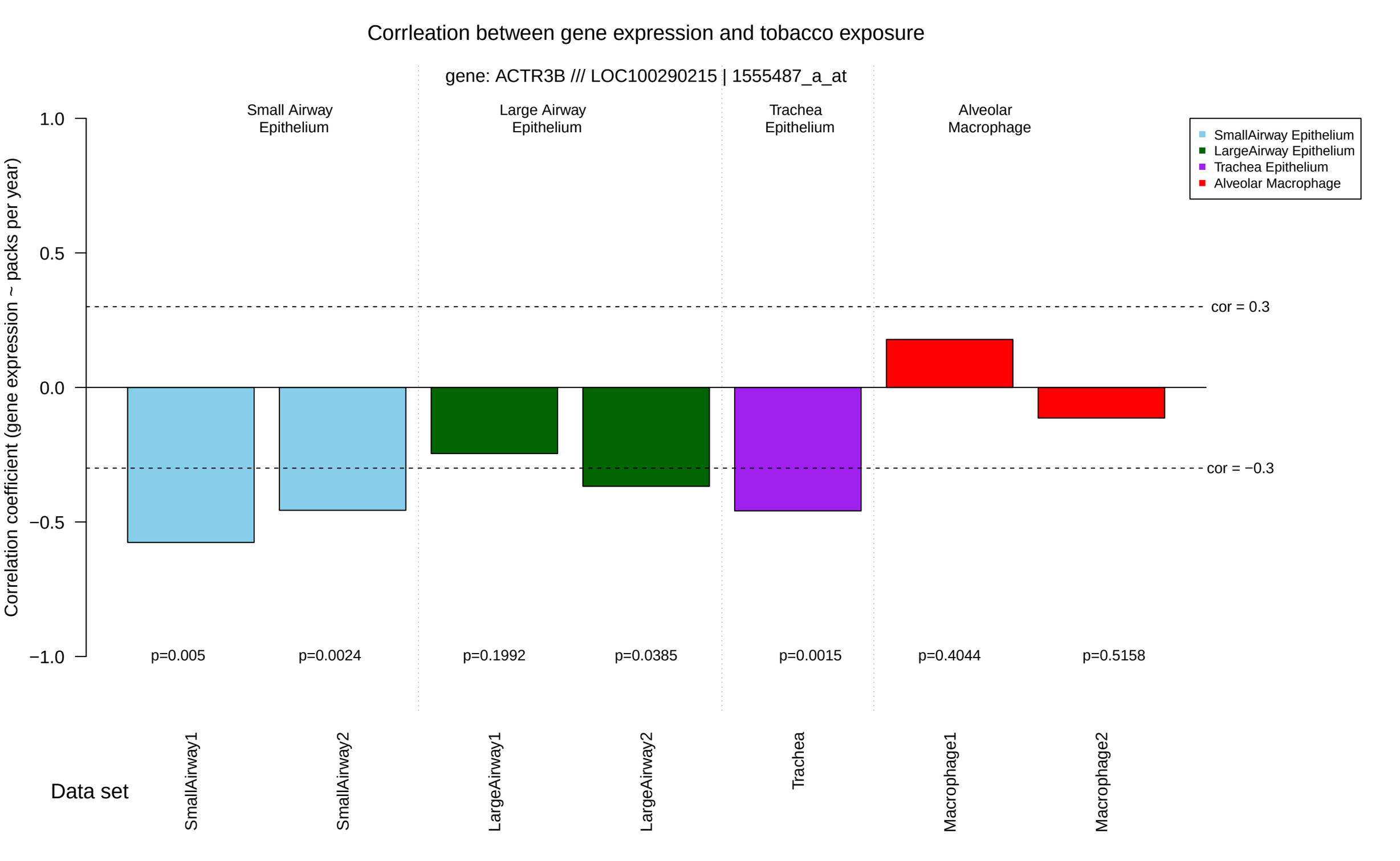

set: LargeAirway1 (p=0.1134)
ACTR3B /// LOC100290215 | 1555487_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0034)
ACTR3B /// LOC100290215 | 1555487_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
ACTR3B /// LOC100290215 | 1555487_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1395)
ACTR3B /// LOC100290215 | 1555487_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5985)
ACTR3B /// LOC100290215 | 1555487_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR3B /// LOC100290215 | 1555487_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.005
p=0.0024
p=0.1992
p=0.0385
p=0.0015
p=0.4044
p=0.5158
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2